Kataram Navya Sree

+91 6304891956 kataramnavyasree05@gmail.com LinkedIn GitHub Portfolio

SUMMARY

Final-year B.Tech CSE (AI & ML) student skilled in Python, Machine Learning, Generative AI, and backend devel opment. Experienced in developing LLM security frameworks, predictive analytics, and time-series forecasting using FastAPI, Scikit-learn, LangChain, and ARIMA. Strong foundation in data preprocessing, model evaluation, REST API development, and building reliable, scalable AI applications.

SKILLS

Programming Languages	Python, Java, SQL, HTML, CSS
AI / Generative AI	LangChain, Prompt Injection Detection, LLM Security, RAG, NLP
Machine Learning	Scikit-learn, Random Forest, LSTM, ARIMA, Time-Series Forecasting
Backend & APIs	FastAPI, REST APIs, Streamlit
Data & Visualization	Pandas, NumPy, Matplotlib, Feature Engineering, Data Preprocessing
Databases	MySQL, SQLite
Developer Tools	Git, GitHub, VS Code, Google Colab
Core Concepts	Model Evaluation, Risk Scoring, Attack Classification, Data Structures and Algo- rithms
Soft Skills	Analytical Problem-Solving, Team Collaboration, Self-Motivated, Communication, Critical Thinking, Organizational Skills

EXPERIENCE

Project Lead — Academic Projects 2023 - 2026
Sreyas Institute of Engineering and Technology Hyderabad, India

- Developed three AI/ML projects in LLM security, job analytics, and air quality forecasting.
- Built AI solutions using Python, FastAPI, Streamlit, Scikit-learn, ARIMA, and LangChain.
- Managed data preprocessing, model evaluation, backend integration, testing, and GitHub version control.
- Improved project delivery through task management, collaboration, and systematic testing.

PROJECTS

Robust LLM Framework for Detecting Prompt Injection and Jailbreak Attacks *(Python, FastAPI, Scikit-learn, LangChain, Prompt Injection Detection, LLM Security)*: June 2026 - Ongoing

- Developed a Python-based LLM security framework using FastAPI, Streamlit, Scikit-learn, and LangChain to detect prompt injection, jailbreak attempts, and malicious instructions in documents and RAG pipelines.
- Implemented risk scoring, attack classification, explainable threat detection, input sanitization, and allow/block mitigation to strengthen LLM application security using free and open-source technologies. (GitHub)

AI-Powered Job Market Analytics & Skill Forecasting Dashboard *(Python, Random Forest, ARIMA, FASTAPI, Resume Analysis)*: December 2025 - April 2026

- Designed and tested a forecasting platform processing 2,000+ job records, combining Random Forest (role predic tion) and ARIMA (skill-demand forecasting) models.
- Built and validated an NLP-based Resume Analyzer that cut manual resume-screening effort by 60%, serving predictions to a React dashboard through a scalable FastAPI backend. (GitHub)

Air Quality Prediction Using Machine Learning *(Python, LSTM, ARIMA, Pandas, NumPy, Time-Series Forecasting)*: January 2025 - June 2025

- Developed test strategies to benchmark LSTM against ARIMA, delivering 88% accuracy on 30-day AQI forecasts.
- Owned the end-to-end pipeline solo-preprocessing, feature engineering, training, and visualization. (GitHub)

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning Expected 2027
Sreyas Institute of Engineering and Technology — CGPA: 7.3/10

CERTIFICATIONS & ACHIEVEMENTS

- AI/ML for Geodata Analytics – Indian Institute of Remote Sensing (IIRS), ISRO, Dept. of Space
- Artificial Intelligence Fundamentals – IBM
- Principles of Generative AI & Deep Learning for Developers – Infosys Springboard
- Software Engineer Certification – HackerRank
- Hackathon Participant – India AI Impact Summit 2026 Pre-Summit Event
- 5-Day Hands-On Generative AI Workshop – Innomatics Research Labs